Maths Exploring for Terminal-06 Solutions

Corrected MCQs

1. **C.** Three independent positions of four choices, including zero at the beginning: $4^3 = 64$.
2. **B.** $5 \times 4 = 20$. The order of duties counts and the same person does not cumulate both.
3. **D.** $4! = 4 \times 3 \times 2 \times 1 = 24$.
4. **A.** Each element is counted once. Without disjunction, the intersection should be removed.
5. **C.** The list keeps the order and allows to reuse a card; there are 2^2 = Four exits.
6. **B.** The group is not ordered, so $\binom{8}{3} = 56$; the ordered product counts each group 3 ! Once.
7. **D.** For each element, two decisions: retained or not. $5^2 = 32$, including the empty part.
8. **A.** Separate groups containing a fixed element (choose two out of five) and those containing a fixed element Not containing (three out of five).
9. **C.** $\binom{5}{2} - \binom{2}{2} = 10 - 1 = 9$. Other calculation: $3 \times 2 + \binom{3}{2} = 9$. Six only count mixed pairs.
10. **B.** Selecting the three members of Group 1 determines Group 2: $\binom{6}{3} = 20$. Divide by 2 correct only for unnumbered groups.

True/false. False: there is an empty part, so $\binom{n}{0} = 1$. True: Question 9. False: A simultaneous draw gives a part; add an order changes the object counted.

Corrected applications and support

Application 1. Codes: 01,02,10,12,20,21. The last four only represent numbers to
A code can start with zero.

Application 2. Two red pieces: one outfit. Exactly one: 1 $\times 2 + 3 \times 1 = 5$ outfits. Both
cases are disjunct.

Implementation 3. Exactly two finals: $\binom{6}{2}\binom{4}{1} = 60$. Joint group: $\binom{10}{3} - \binom{6}{3} - \binom{4}{3} = 96$; or
$\binom{6}{2}\binom{4}{1} + \binom{6}{1}\binom{4}{2} = 60 + 36 = 96$.

Implementation 4. Each unordered pair is counted twice; there is $\binom{5}{2} = 10$ pairs.
roles distinguish the choices, hence 20.

Application 5. Line : 1, 5, 10, 10, 5, 1, sum 32. The first 10 count the parts of size 2 d
size 5 set; the second size 3 set.

Implementation 6. Two zeros: $\binom{4}{2}4^2 = 96$. Of the five figures, three are even and two are odd.
odd: $\binom{4}{1}2\,3^3 + \binom{4}{3}2^3\,3 = 216 + 96 = 312$.

Institutionalization

Ask before each calculation: "What is a possible result? When are two results
allow lists and trees in the small case, then make explicit what becomes general.
The calculation of parity can remain a challenge accompanied; the case-by-case method suffices.

Conduct and correction

Benchmarks for the teacher

The common duty S28 precedes the combinations of S30: to put only principles and lists
The DS3 can mobilize combinations, but not Pascal's relationship taught in
S35. Make the object counted before any formula. For fragile students, use cards
with names; does a card swap change the object? The Newton binome, quoted
in the progression, is a deepening: keep it explicitly optional.

Corrected activity

1. A binomial is a part of two of six pupils: $\binom{6}{2} = \frac{6\times5}{2} = 15$.

2. The president can be chosen in six ways, then the secretary from among the other five students: 6 $\times 5 = 30$ distributions.

3. Each intervention may be entrusted to one of the six students, including one already chosen student. three successive choices give 6 $^3 = 216$ sequences.

4. With four cards, the pairs are AB, AC, AD, BC, BD, CD : six choices. The twelve couples of roles are:

$$\begin{array}{lll} (A,B) & (A,C) & (A,D) \\ (B,A) & (B,C) & (B,D) \\ (C,A) & (C,B) & (C,D) \\ (D,A) & (D,B) & (D,C) \end{array}$$

   There are three follow-up interventions: $^3 = 64$, for each position has four choice.

5. For a pair, the order doesn't count and you don't repeat a student. For roles, the order counts, For interventions, order counts and repetition is allowed.

E1. There are 10^4 Those with no 0 have nine choices for each of the four Positions 9 ^{4} codes. As a complement, the requested number is 10 $^4 - 9^4 = 3439$.

E2. There are ten choices for gold, then nine for silver and eight for bronze. gives 10 $\times 9 \times 8 = 720$ podiums. Changing order changes the podium.

E3. The person imposed is already chosen. It remains to choose a part of two people from the Six others: $\binom{6}{2} = \frac{6\times5}{2} = 15$.

E4. A path is determined by the positions of its four steps to the right among seven positions. Three more steps are then imposed. $\binom{7}{4} = 35$ paths.

E5. You have to choose two girls out of eight, then two boys out of seven. gives only one group: $\binom{8}{2}\binom{7}{2} = 28 \times 21 = 588$.

Corrected from DM9

1. Order does not count: there is $\binom{12}{4} = \frac{12\times11\times10\times9}{24} = 495$ commissions.

2. Two students are selected from among seven and two teachers: $\binom{7}{2}\binom{5}{2} = 21 \times 10 = 210$ commissions.

3. The 495 commissions were withdrawn. $\binom{5}{4} = 5$ commissions exclusively teachers and $\binom{7}{4} = 35$ These categories are disjunct: 495 $- 5 - 35 = 455.$

4. Each of the 210 commissions has two choices of teacher president and two choices of student secretary: $210 \times 2 \times 2 = 840$ organisations.

5. Classify commissions by number k teachers, with $0 \leq k \leq 4$. Category k contains $\binom{5}{k}\binom{7}{4-k}$ The five categories are disjunct and cover all commissions. addition,

$$\sum_{k=0}^{4} \binom{5}{k}\binom{7}{4-k} = \binom{12}{4} = 495.$$

Corrected work in the manual or an anal

One hand contains five distinct cards; the order does not intervene. $\binom{52}{5} = 2{,}598{,}960$ hands. For a hand containing the spades, this card is already chosen; the other four are chosen from among The remaining 51 cards. There are $\binom{51}{4} = 249{,}900$ hands. This result is not multiplied by 5 : the hand n no first or last position.